CSE221: Algorithms
Lab-01
Summer 2026

Action required for Students

[Google Form] CSE221 Lab Policy:
- The lab policy file can be found here - Link
- Please complete the lab policy acknowledgement form - Link

[Google Form] Codeforces Account Information: (Action required)
- Open an account at https://codeforces.com/
- Please submit your codeforces details here - Link

Students who do not complete this will not be allowed to participate in lab quizzes, and their assignment submissions will not be graded.

Necessary Resources [Bookmark the Link]

Student Drive Folder: Link
The folder contains:
- Lab Policy
- Sample Quiz Checkers
- Environment Setup for Quiz Checker
- Previous Semesters' Lab Quiz Questions

Assignment-01 Links:

codeforces Invitation Link:
https://codeforces.com/contestInvitation/a690bb31bdd271a6834978c6c19324e617e249b1

Alternative Link: https://codeforces.com/contests/699310
[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: `Jun 28, 2026 2:00 PM`

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

Learning Outcome of this Lab:

- Understand the basics of Online Judge platforms.
- Become familiar with standard Input/Output (I/O) formats used in Online judge
- Learn about common judge verdicts, including Accepted (AC), Wrong Answer (WA), Time Limit Exceeded (TLE), and Runtime Error (RE).
    - For a **1-second** time limit, most OJs allow approximately **10^8 operations**.
    - For python select PyPy over Python if available, as PyPy is faster. However, PyPy consumes more memory than Python.
- Review $O(n^2)$ sorting algorithms which have already been covered in the previous courses such as Bubble Sort, Selection Sort, and Insertion Sort.
    - Learn to sort with multiple parameter
    - Understand the use case of stable sort
- Explore Fast I/O techniques (**self-study**):
    - **Python**: Using `sys.stdin` for faster input handling.
    - **C++**: Using `ios_base::sync_with_stdio(false);` and `cin.tie(nullptr)`
    - **Java**: Using `BufferedReader` for efficient input reading.
    - The fast i/o can reduce program execution time on Online Judge platforms.
    - **Fast I/O Example: Python, C++, Java [click Expand to view the code snippet]**